Real-Time Convex Combination of Adaptive Filters Using Mobile Devices

Vishrut Sharma(25210157)
Vikash Kumar Sinha(25210156)

April 11, 2026

1 Introduction

Adaptive filtering is widely used in system identification, noise cancellation, and communication systems. In this work, we implement a real-time adaptive filtering framework using two filters:

- Normalized Least Mean Squares (NLMS)
- Trigonometric Functional Link Artificial Neural Network (TFLANN)

A convex combination of these filters is used to exploit the fast convergence of NLMS and the nonlinear modeling capability of TFLANN.

2 System Overview

The system consists of two mobile devices:

- **Phone A**: Captures real-time audio input via microphone
- **Phone B**: Receives input via TCP/IP and performs adaptive filtering

The received signal is processed through NLMS, TFLANN, and their convex combination.

3 Convex Combination Model

The combined output is given by:

$$y(n) = \lambda(n)y_1(n) + (1 - \lambda(n))y_2(n) \tag{1}$$

where:

- $y_1(n)$: NLMS output
- $y_2(n)$: TFLANN output
- $\lambda(n)$: adaptive mixing parameter

The mixing parameter is defined using a logistic function:

$$\lambda(n) = \frac{1}{1 + e^{-\alpha(n)}} \tag{2}$$

The update rule for $\alpha(n)$ is:

$$\alpha(n+1) = \alpha(n) + \mu_\lambda \cdot e(n) \cdot (y_1(n) - y_2(n)) \cdot \lambda(n)(1 - \lambda(n)) \tag{3}$$

4 Implementation in Simulink

The complete system is implemented using block-based modeling. The major components include:

4.1 Input Acquisition

Audio input is captured on Phone A and transmitted to Phone B using TCP/IP blocks.

4.2 Adaptive Filters

- NLMS filter for linear modeling
- TFLANN with trigonometric expansion for nonlinear modeling

4.3 Convex Combination

A weighted sum of both filter outputs is computed using adaptive $\lambda(n)$.

4.4 Error and MSE Calculation

The error signal is:

$$e(n) = d(n) - y(n) \tag{4}$$

Mean Square Error (MSE) in dB is computed as:

$$\text{MSE} * dB = 10 \log * 10 \left(\mathbb{E}[e^2(n)] \right) \tag{5}$$

5 Experimental Setup

- Input: Real-time microphone signal
- Communication: TCP/IP between mobile devices
- Platform: Simulink deployed on Android device
- Filters:
  - NLMS (fast convergence)
  - TFLANN (nonlinear modeling)

6 Results

6.1 Convergence Behavior

- NLMS converges quickly but struggles with nonlinearities
- TFLANN converges slower but achieves lower steady-state error
- Convex combination achieves best performance overall

6.2 Lambda Evolution

- $\lambda(n)$ starts at zero (NLMS for fast convergence)
- Gradually adapts based on filter performance
- Stabilizes at one (TFLANN for lower MSE)